Spectral Harmonization Landsat-8 and Sentinel-2A: The Matching Bands Adjustment Before the Missing Bands Prediction Method

Aojie Shen and Jie Shao

***Abstract*— The Sentinel-2 MSI and Landsat-8 OLI are often used as data sources to form temporally dense harmonization observations for accurate quantitative studies. However, challenges remain in resolving surface reflectance differences between both sensors due to inconsistencies in spectral band positions, widths, and spectral response functions. In this study, we developed the matching bands adjustment before missing bands prediction method (MA-MP) for performing spectral harmonization between Landsat-8 and Sentinel-2A to reduce such surface reflectance differences. This method uses the global representative samples from the harmonized Landsat Sentinel-2 (HLS) product to build the spectral harmonization model of Landsat-8 and Sentinel-2A at the cluster level for band-pass adjustment in the bands where Landsat-8 and Sentinel-2A are matched, and for prediction in the band which is unilaterally missing bands in Landsat-8 image relative to Sentinel-2A image. Experimental results show robust spectral harmonization in four selected study areas with a variety of land cover types, and the spectral harmonization results are significantly better than those of current mainstream methods. Overall, the proposed MA-MP can adjust the observations of Landsat-8 and Sentinel-2A in matching bands and predict the observations of Landsat-8 in a unilaterally missing band, which has the potential to be used flexibly for building the spectral harmonization model between any two different sensor images.**

***Index Terms*— Band-pass adjustment, HLS, Landsat-8, missing band prediction, Sentinel-2A, spectral harmonization.**

I. Introduction

SATELLITE data with high spatiotemporal resolution are highly desired for studying the dynamics of our Earth systems [1], [2]. However, it is difficult for single-satellite remote sensing data to balance high spatial resolution and high temporal resolution for technical and financial constraint reasons [3], [4]. Most satellites with a coarse spatial resolution (250–1000 m) have a high revisit frequency. For example, MODIS acquired images with a coarse spatial resolution (250 m in red and Nir bands and 500 m in other land reflectance bands) but with a high temporal resolution (daily) [5]. Geostationary earth orbit (GEO) satellites generally have high temporal/low spatial resolution [6], such as GOES-R has the temporal resolution of 15 min but the spatial resolution of 2 km [7]. This category of data is not ideal for detecting the land surface change that is usually occurring at a smaller scale than their pixel sizes [8], [9]. In contrast, medium-resolution satellites such as Landsat can acquire relatively high spatial resolution images (30 m) [10] and provide fine enough observations to monitor a variety of landscapes [11], [12]. However, Landsat observations have a low temporal resolution (a revisit period of 16 days). The low frequency of observations means that even with two similar sensors working at the same time (such as Landsat-7 and Landsat-8), the same place can only be visited again for nearly a week. As a result, it is often difficult to capture information about critical periods by using such remote sensing images [13]. Moreover, cloud contamination during passing time masks the desired information further, resulting in decreases in valid observations [14].

With the development of global Earth observation techniques, an increasing number of satellites are being developed and operating in orbit [15]. Therefore, it is possible to combine remote sensing images from multiple satellites to improve the observation frequency [16] and hence to increase the chance of cloud-free acquisitions [17]. However, combining multiple satellite images is a very challenging task, mainly due to different acquisition geometries or illumination conditions, spatial resolutions, spectral band positions and widths, and spectral response functions [18], [19]. The difference introduced by different acquisition geometries or illumination conditions can be reduced and solved by the bidirectional reflectance distribution function (BRDF) correction [20], [21]. Meanwhile, image resampling can adjust the spatial resolution of one sensor to the other [22]. However, the reflectance differences caused by different spectral band positions, widths, and spectral response functions (hereafter referred to as "spectral difference") are still challenging for the remote sensing community [23].

In general, even if spectral bands with similar wavelength positions exist (hereafter referred to as the "matching band"), the differences in the spectral response functions between different sensors will create slightly different observations. This will cause further variations in the spectral index values

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or classification maps, leading to misinterpretation of the results [24]. Additionally, there will be unilaterally missing bands (hereafter referred to as the "missing band") in one sensor image relative to another for the purpose of satellite design and the budgetary, and/or technical constraints [25]. These missing bands often carry important information. For example, compared to the Landsat-8 OLI, the Sentinel-2 MSI offers three extra red-edge bands [26] that are valuable in vegetation leaf chlorophyll estimation and biotic stresses detection [27], [28]. To effectively reduce the spectral difference, it is necessary to implement solutions in the above two hands—the process is referred to as "spectral harmonization." Among spectral harmonizations, adjusting the spectral differences between the matching bands is called band-pass adjustment [29], and the process for solving the problem of missing band is called missing band prediction [30].

Currently, a series of band-pass adjustment algorithms are proposed [31], [32]. Based on Sentinel-2 MSI and Landsat-8 OLI surface reflectance values simulated from the Hyperion dataset on Earth Observing-1 [33], NASA scientists built global fixed per-band linear regression (LR) models to adjust the spectral differences of matching bands between Landsat-8 and Sentinel-2 observations, and further combined with other processing processes (such as atmospheric correction, geometric resampling, geographic registration, and BRDF correction) to form the harmonized Landsat and Sentinel-2 (HLS) surface reflectance product [29]. There are studies that similarly generated regional fixed per-band LR models in Australia [34] and the conterminous United States [35]. The fixed per-band band-pass adjustment models are convenient but not applicable to various scenes at the regional or global scale [35]. Although using a global band-pass adjustment model can reduce the spectral difference of matching bands, large spectral differences still exist between different sensor observations [23].

A time series-based reflectance adjustment approach (referred to as "TRA") was proposed for improving band-pass adjustment accuracy, which built pixel-by-pixel band-pass adjustment model instead of a global band-pass adjustment model [23]. The results of the study showed that band-pass adjustment based on TRA reduced the spectral difference between Landsat-8 and Sentinel-2 surface reflectance in the matching bands and outperformed the HLS product. However, compared with global or regional band-pass adjustment models, the TRA approach is time-consuming for building the regression model at the pixel level. Additionally, TRA does not take into account the possible pixel changes in time, so the built band-pass adjustment models will no longer be applicable when the land cover types of the pixels are changed.

To balance band-pass adjustment accuracy and computational efficiency, Scheffler et al. [30] implemented band-pass adjustment by using land cover optimized multivariate regressors. They clustered the simulated Landsat-8 and Sentinel-2A data by convolving sensor spectral response functions and airborne hyperspectral data [36] and built multivariate regression models for band-pass adjustment at the cluster level. Owing to the spectral characteristics of different landscape properties, the performance of the band-pass adjustment model obtained at the cluster level was better than that of the global band-pass adjustment model. This suggests that different landscape properties should use separate band-pass adjustment models for band-pass adjustment. Moreover, the band-pass adjustment model built at the cluster level obviously requires less time than the model built at the pixel level. However, hyperspectral images usually suffer from spatial and temporal discontinuities, resulting in missing images at specific time periods and geographic areas, so it is difficult to obtain sufficient representative samples for building band-pass adjustment models [37]. Therefore, based on samples from hyperspectral images, it is difficult to build band-pass adjustment models that can be applied to various scenes.

In conclusion, none of the mentioned studies can build band-pass adjustment models that can account for the different landscape properties of pixels while being universally applicable to various scenes.

Currently, researchers tend to consider only the multisource remote sensing images with matching bands or even discard remote sensing images with missing bands, resulting in less attention to missing band prediction. Scheffler et al. [30] developed a method to build a multivariate linear regression (MLR) model between the observations of Landsat-8 OLI images in the bands matched to Sentinel-2 MSI images and the observations of Sentinel-2 MSI images in the red-edge band at the cluster level and predicted the observations of Landsat-8 OLI images in the red-edge band based on the model built. They further applied the predicted red-edge band data to the estimation of burn severity [38] and obtained robust estimation results. However, the original spectral differences between Landsat-8 OLI and Sentinel-2 MSI observations are passed on to the missing band prediction model built by Scheffler et al. [30] and lead to reduced prediction accuracy. Zhou et al. [25] built a missing band prediction model based on selected similar pixels considered spatially heterogeneous cover types of the pixels. This approach is time-consuming, as building the missing band prediction model at the pixel level. The acquisition of similar pixels requires high-quality observation images, which are difficult to meet universally. Therefore, it is necessary to develop new methods for building a missing band prediction model that has high prediction accuracy and can be universally applied to various scenes. In addition, when performing spectral harmonization, current studies tend to focus only on band-pass adjustment or missing band prediction, such spectral harmonization is not complete.

To address the challenge mentioned above, this study proposes the matching bands adjustment before the missing bands prediction method (MA-MP) for performing spectral harmonization between Landsat-8 and Sentinel-2A, where MA-MP is a complete spectral harmonization method that includes both band-pass adjustment and missing band prediction. Specifically, MA-MP uses global representative samples to build the spectral harmonization model of Landsat-8 and Sentinel-2A at the cluster level, which aims to enhance the universal applicability of the spectral harmonization model across various scenes. Furthermore, the MA-MP developed a new method for building missing band prediction models based on the spectral curve characteristics of the pixels with different landscape properties for improving missing band prediction accuracy. This article is structured as follows. First, we introduce the satellite data and study areas, and then the

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MA-MP is described. Next, we quantitatively evaluate the spectral harmonization results from the MA-MP in band-pass adjustment and missing band prediction, respectively. Then, we discuss the results. This is followed by the conclusion, including implications and recommendations.

II. Data and Study Areas

A. Datasets

Sentinel-2 MSI can provide "Landsat-like" observations because it has similar band designs as Landsat-8 OLI [26] in visible (blue, green, and red), near-infrared (Nir), and two short-wave infrared (SWIR1 and SWIR2) bands, and it has the red-edge band which is lacking in Landsat-8 OLI. Therefore, Sentinel-2 MSI data and Landsat-8 OLI data are often used as data sources to form temporally dense harmonization observations [39], [40]. In this study, we selected Landsat-8 and Sentinel-2A images from 2016 to evaluate the MA-MP.

To avoid the surface reflectance differences caused by different acquisition geometries, illumination conditions, and atmospheric states between Landsat-8 and Sentinel-2A observations [18], [41], [42], the HLS v1.4 product (hereafter referred to as "HLS product") [29] was used as the data source. The HLS product was produced by harmonizing images of Sentinel-2 MSI and Landsat-8 OLI [43] with five harmonization processes: atmospheric correction [44], [45], cloud masking [46], coregistration [47], view and illumination angle normalization [20], [48], and band-pass adjustment [29]. Among these processes, the band-pass adjustment in the HLS product was carried out using the fixed per-band LR model at the global scale. These harmonization processes give HLS data global observations with a spatial resolution of 30 m, a temporal resolution of every 2–3 days, and the military grid reference system (MGRS) in the Universal Transverse Mercator (UTM) projection [49], [50].

We referred to HLS Landsat-8 and HLS Sentinel-2A as L30 and S30, respectively, and the BRDF-corrected Sentinel-2A derived by reversing the HLS band-pass adjustment from the S30 surface reflectance based on the fixed per-band regression coefficients provided in the HLS product was referred to as NadirS30 [23]. The detailed band numbers and band-pass adjustment coefficients of Landsat-8 and Sentinel-2A are shown in Table I (from the product user's guide, https://hls.gsfc.nasa.gov/documents/). This study performed spectral harmonization for converting L30 surface reflectance to NadirS30 surface reflectance. The HLS product does not perform BRDF correction on the red-edge bands of Sentinel-2A images, so only the HLS product in the visible, Nir, SWIR1, and SWIR2 bands were used for spectral harmonization in this study.

In the HLS product, the QA band provides information about aerosol quality, water, snow/ice, cloud shadow, adjacent cloud, cloud, and cirrus for each pixel. The detailed description of the product is referred to the product user's guide (https://hls.gsfc.nasa.gov/documents/, accessed on 6 September 2022). We first masked out pixels that were indicated as highly impacted by aerosol in the HLS images (the bit combination of bits 7-6 is 11 in the QA image). Then, we masked out pixels indicated as cloud shadow, adjacent cloud, cloud, and cirrus. The remaining pixels were called clear pixels.

TABLE I
Band Numbers and Band-Pass Adjustment Coefficients of the HLS Products Used in This Study

Band Name	L30 Band Number	NadirS30 Band Number	Slope	Intercept
Blue	2	2	0.9778	-0.004
Green	3	3	1.0053	-0.0009
Red	4	4	0.9765	0.0009
NIR	5	8A	0.9983	-0.0001
SWIR1	6	11	0.9987	-0.0011
SWIR2	7	12	1.003	-0.0012

We selected finer resolution observation and monitoring of global land cover (FROM-GLC) to provide land cover information for the study area. FROM-GLC is the first 30-m resolution global land cover map produced using Landsat thematic mapper (TM) and enhanced thematic mapper plus (ETM+) data [51]. By manually interpreting TM/ETM+ images, 91 433 training samples and 38 664 test samples can be obtained, and based on these samples, FROM-GLC can be produced [52]. The quality of the two sample sets has high certainty because most of these samples were interpreted using ancillary high-resolution viewable imagery from Google Earth. The interpretation was primarily carried out based on a color composite of TM images of Bands 4, 3, and 2 displayed with red, green, and blue, respectively [53]. Data downloads and specific classification systems can be found at http://data.starcloud.pcl.ac.cn/zh/resource/2. In this study, we selected classified images from 2017 to provide information on the land cover types of the pixels.

B. Study Area

Four HLS tiles (each HLS tile covers 109.8 × 109.8 km) were selected for spectral harmonization according to the land cover types, the qualities of the L30 and NadirS30 data, and the observation time interval between the L30 and NadirS30 data (see Fig. 1 and Table II). Note that the main land cover types mean that the percentage of pixels belonging to these land cover types exceeds 3% in the study areas.

Specifically, the LB site (HLS ID: 34RGS) is mainly covered by bareland. The land cover type at the SP site (HLS ID: 29SQB) is mainly cropland, followed by forest and shrubland, and contains a little impervious surface and water. The land cover types at the ZB (HLS ID: 34LGJ) and the PG sites (HLS ID: 21JWM) are mainly grassland, followed by shrubland and forest at the ZB site and forest and cropland at the PG site, respectively. In addition to the main of land cover types, there is approximately 1% wetland at the ZB site. The four sites contained 80% of the land cover types in the classification system of the FROM-GLC products (except for Tundra and Snow/Ice), which indicates that the land cover types at these selected sites are sufficiently representative. The main land cover types at each site are shown in Table II, and the standard false-color-composite L30 images made out of the Nir, red, and green bands are shown in Fig. 2.

Considering the effect of cloud cover contamination on image quality, only the images with more than 70% clear pixels were selected from the four sites. Additionally, we needed

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TABLE II
DETAILED INFORMATION ABOUT STUDY AREAS

Location	Site name	HLS ID	Center latitude and longitude	NadirS30 DOY	L30 DOY	Main land cover types (%)
Paraguay	PG	21JWM	(25°46′S, 56°26′W)	119	120	Grassland (66%), Forest (26%) and Cropland (5%)
Spain	SP	29SQB	(37°25′N, 06°08′W)	358	358	Cropland (63%), Forest (18%), Shrubland (9%), Water (4%), and Impervious surface (4%)
Zambia	ZB	34LGJ	(14°56′S, 23°23′E)	137	137	Grassland (48%), Shrubland (38%), and Forest (10%)
Libya	LB	34RGS	(28°16′N, 23°3′E)	309	307	Bareland (99%)

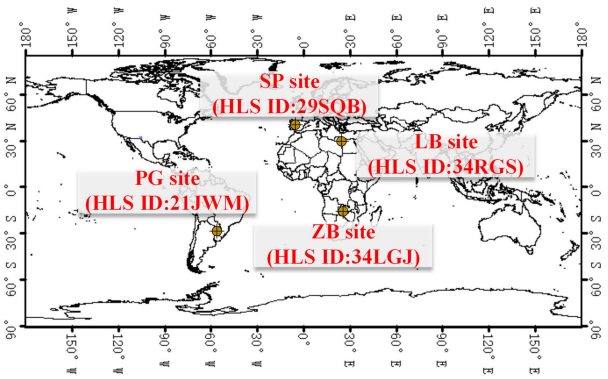


Fig. 1. HLS IDs and locations in four HLS tiles.

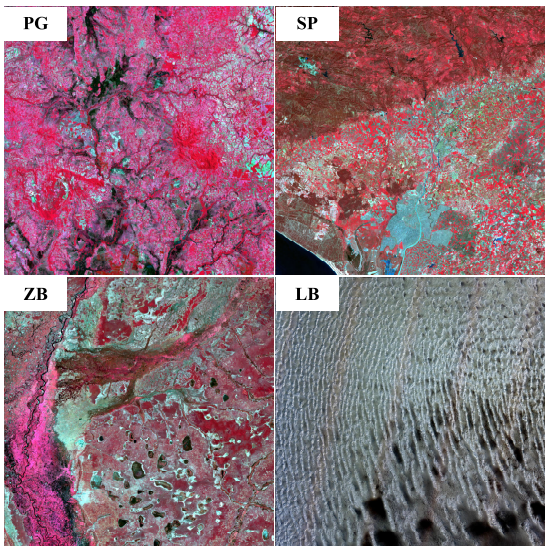


Fig. 2. Standard false-color-composite L30 images in the four sites (the site names were in the upper left corner).

to avoid spectral differences between L30 and NadirS30 data caused by larger time intervals. Studies show that data collected with a time interval of less than one day should have similar values [23], so we generally used L30 observations collected within ±1 day of the NadirS30 acquisition date to match the NadirS30 observations. At site LB only with desert landscape (labeled Bareland in the FROM-GLC product), based on image quality, we selected L30 and NadirS30 observations with a two-day time interval (see Table II).

III. METHODS

The MA-MP involves three major steps: 1) constructing the sample library, 2) building the band-pass adjustment model and missing band prediction model at the cluster level, and 3) spectral harmonization for L30 images. The flowchart of MA-MP is shown in Fig. 3.

A. Sample Library Construction

Constructing the sample library involves tile screening, image selecting, and pixel sampling. Tile screening involves screening HLS tiles globally that contain various land cover types to ensure that the pixels in the sample library were spatially representative. In this process, we used FROM-GLC data combined with manual interpretation to make the screened tiles have representative land cover types.

Then, considering that the reflectance of landscapes may change over time (e.g., the reflectance of vegetation in different phenology periods is significantly different), image selecting was required. Specifically, we selected images from different seasons (spring, summer, autumn, and winter) in each of the screened tiles. Considering cloud cover contamination, images with more than 70% clear pixels were mainly selected. To avoid spectral differences caused by the observation time interval between L30 and S30, referring to existing studies [23], we selected images with a time window of less than one day between L30 and S30 observations as the matching image pairs. For partly cloudy periods, we expanded the time window to one week to select matching image pairs.

Finally, the sample library was constructed by pixel sampling from the selected images. Pixel sampling requires two steps: obtaining the available pixel pairs and setting the number of pixel samples. To further avoid sudden land cover changes between the selected image pairs, we used the filter [48] for surface reflectance values of the clear pixel pairs

$$\frac{\left|\rho^{\mathrm{L30}}(\lambda,i)-\rho^{\mathrm{S30}}(\lambda,i)\right|}{\left|\rho^{\mathrm{L30}}(\lambda,i)+\rho^{\mathrm{S30}}(\lambda,i)\right|/2}>0.5 \tag{1}$$

where $\rho^{\mathrm{L30}}(\lambda,i)$ is the L30 surface reflectance, $\rho^{\mathrm{S30}}(\lambda,i)$ is the S30 surface reflectance, λ is the band index, and i is the ith pixel pair of surface reflectance.

We selected quite aggressive filtering thresholds [48] to ensure that the surface reflectance differences of the retained pixel pairs were mainly due to spectral differences and removed pixel pairs that did not satisfy the threshold constraint even in only one band. Such processing resulted in a certain number of pixel pairs being removed. However, given the accuracy of the built spectral harmonization model, it was preferable to remove pixels that contained any change. The corresponding remaining pixels were then identified as the available pixels. We only sampled available pixels from

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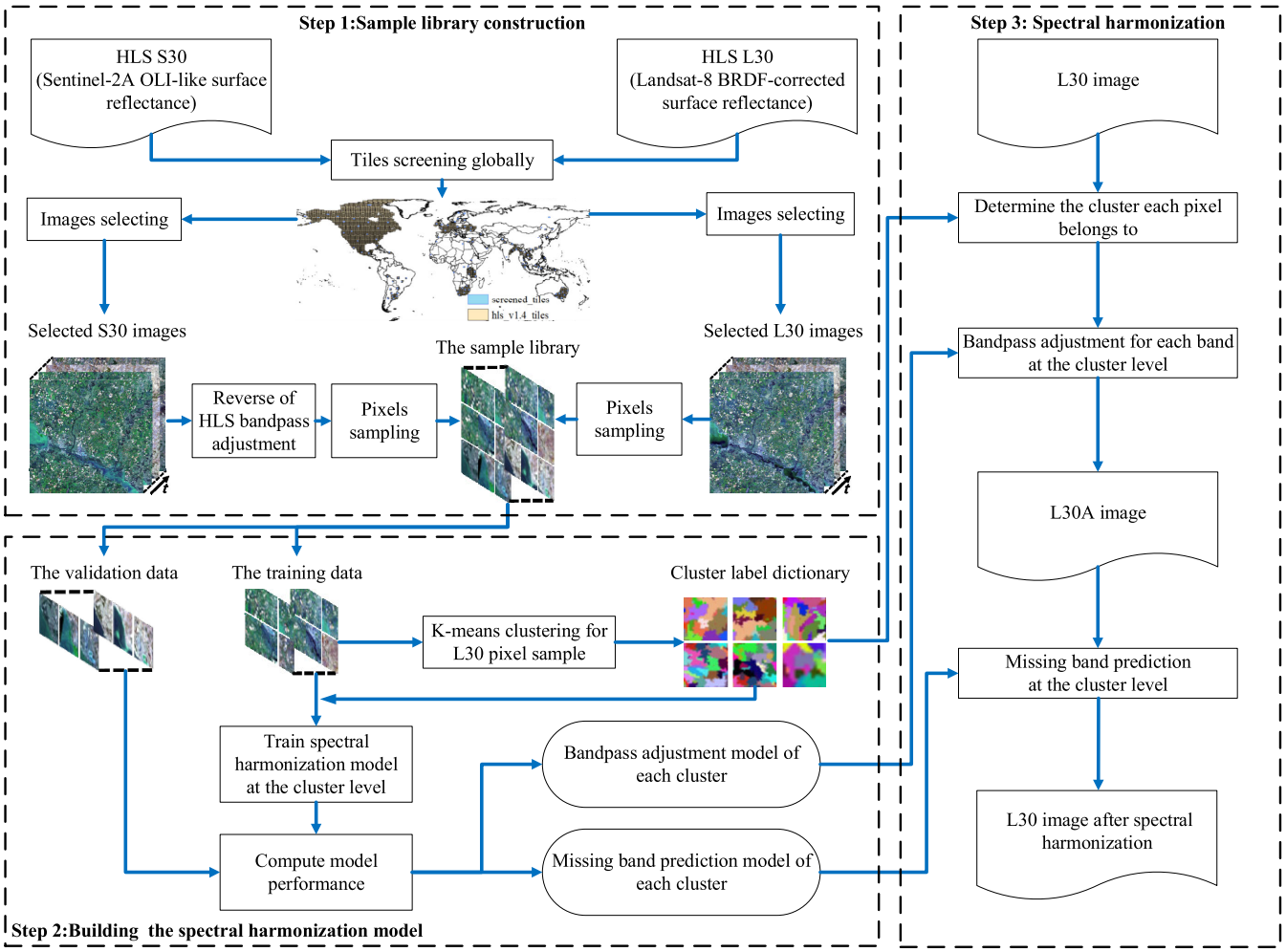


Fig. 3. Flowchart of the MA-MP, involving constructing the globally representative sample library, building the band-pass adjustment model and the missing band prediction model, and the implementation of the spectral harmonization for L30 Images.

selected images to ensure the accuracy of the spectral harmonization model. Then, for available pixels in selected S30 images, we reversed HLS band-pass adjustment to transfer S30 to NadirS30 using the LR model with the fixed per-band regression coefficients from Table I.

To make the random pixel samples reflect the land cover types of the image, we tested different numbers of pixel samples to obtain pixel samples from several HLS images and compared the land cover type percentages between the pixel samples and the HLS images. We empirically increased the number of pixels sampled from 1000 to 9000 pixels gradually. To avoid the uncertainty caused by only one random sampling, we sampled 100 times at each number of pixel samples and took the average of land cover type percentages of 100 times sampled sample as the final statistical results at the number of pixel samples. Based on the statistical results and considering the modeling efficiency, we determined the optimal number of pixel samples. In addition, for some cloudy areas throughout the year, we supplemented the images of tiles screened by using the HLS images in 2019 to ensure that the timing of the selected images would cover all seasons.

B. Building the Band-Pass Adjustment Model

When building the band-pass adjustment model between different sensors, the effect of different landscape properties needs to be accounted for to accurately reduce the spectral difference between different sensors [30]. This study divided the pixel pairs in the sample library into two parts, with one part randomly selected as training data to build the spectral harmonization model and the rest used as validation data for evaluating model performance. We first obtained cluster label dictionary by using the K-means algorithm to cluster L30 pixels in the training data and then built the band-pass adjustment model that adjusted the L30 surface reflectance to NadirS30 at the cluster level. We used the surface reflectance of the six selected matching bands between L30 and NadirS30 as the feature of clusters and assigned each pixel to the corresponding spectral cluster based on the Euclidian distance between features. The key parameters that need to be considered in the K-means algorithm are the number of clusters. This study tested different numbers of clusters. The different number of clusters allowed us to later assess the overall band-pass adjustment performance with different numbers of clusters and allowed us to determine the optimal number of clusters to build the band-pass adjustment model.

The LR model is a simple and efficient model and has been widely used for band-pass adjustment [32]. The study has suggested that the LR model with an intercept is a better option than the model without an intercept [23]. We, therefore, built the band-pass adjustment model based on the

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equation below

$$\rho_{c_j}^{\text{NadirS30}}(\lambda) = a_{c_j}(\lambda) * \rho_{c_j}^{\text{L30}}(\lambda) + b_{c_j}(\lambda) \quad (2)$$

where λ is the band index, c_j is the jth cluster, $\rho_{c_j}^{\text{NadirS30}}(\lambda)$ is the NadirS30 surface reflectance, $\rho_{c_j}^{\text{L30}}(\lambda)$ is the L30 surface reflectance, and $a_{c_j}(\lambda)$ and $b_{c_j}(\lambda)$ are the estimated model parameters. By minimizing the difference between the matched NadirS30 and L30 surface reflectance after band-pass adjustment, the model parameters for each cluster pixel were obtained, i.e., to separately fit the band-pass adjustment model for each cluster.

C. Building the Missing Band Prediction Model

We propose a new missing band prediction method based on the spectral curve characteristics of the pixels with different landscape properties. The pixels of each landscape property have unique spectral curve characteristics; therefore, we assumed that the surface reflectance relationship of one band to the other bands was different for pixels with different landscape properties. Based on this assumption, in the NadirS30 samples used as the reference, we built the regression relationship at the cluster level between the surface reflectance of the band to be predicted and the remaining bands matching with L30. And then the L30 observations in these matched bands are brought into the regression relationship of the corresponding cluster to obtain the predicted band's surface reflectance. We evaluated the performance of the MLR model and the random forest regression (RFR) model in missing band prediction for determining the optimal modeling method.

The surface reflectance difference between L30 and NadirS30 in the matching bands was further reduced after band-pass adjustment, at which point it was assumed that the spectral curve characteristics of the same landscape properties of pixels were similar in S30 and NadirS30. Therefore, in the framework of the MA-MP, we used the missing band prediction model built in NadirS30 to the band-pass adjustment L30 (hereafter referred to as L30A) image to obtain the missing band image in L30.

D. Spectral Harmonization for L30 Images

After the spectral harmonization model was built for each cluster, spectral harmonization was applied for each available pixel in the L30 images.

Specifically, in the L30 images, the cluster of each available pixel was first determined based on the surface reflectance of the six matching bands. Then, spectral adjustment of each available pixel was performed based on the band-pass adjustment model corresponding to the cluster to which the pixel belonged. Finally, in the obtained L30A image, the band prediction of each available pixel was performed based on the missing band prediction model corresponding to the cluster to which the pixel belonged.

For each pixel in the L30 image, the spectral harmonization models were determined based on the cluster to which the pixel belonged, rather than using the all spectral harmonization models built based on the constructed sample library. Specifically, if there was only one land cover type in the L30 images, pixels in the image may choose the same spectral harmonization model.

E. Evaluating Model Performance

We selected the mean absolute percentage difference (MAPD) as the metric to quantitatively evaluate the spectral difference between NadirS30 and L30 observations both before and after spectral harmonization and selected the mean relative percentage difference (MRPD) to see over/underestimation [54], [55]. Only the available pixel observations were used for evaluation. The MAPD and MRPD were calculated based on (3) and (4), respectively

$$\text{MAPD} = \left(\frac{1}{N} \sum_{i=1}^{N} \left| \frac{\rho^{\text{NadirS30}}(\lambda, i) - \rho^{\text{L30}}(\lambda, i)}{\rho^{\text{NadirS30}}(\lambda, i)} \right| \right) * 100 \quad (3)$$

$$\text{MRPD} = \left(\frac{1}{N} \sum_{i=1}^{N} \frac{\rho^{\text{L30}}(\lambda, i) - \rho^{\text{NadirS30}}(\lambda, i)}{\rho^{\text{NadirS30}}(\lambda, i)} \right) * 100 \quad (4)$$

where $\rho^{\text{NadirS30}}(\lambda, i)$ is the NadirS30 surface reflectance, $\rho^{\text{L30}}(\lambda, i)$ is the L30 surface reflectance before or after the spectral harmonization, λ is the band index, i is the ith pixel pair of surface reflectance, and N is the total counts of pixel pairs.

In this study, the spectral differences between NadirS30 observations and L30 observations before and after spectral harmonization were quantitatively evaluated. And the results of the spectral harmonization based on the spectral harmonization model from the MA-MP were quantitatively compared with that of the band-pass adjustment based on the global LR model provided by the HLS product (the method is hereafter referred to as "GLR") and that of the missing band prediction based only on MP, which was a prediction based on L30 instead of L30A.

IV. Results

A. Sample Library

By using the method described in Section III-A, we finally screened 112 HLS tiles globally (see Fig. 4), where approximately 500 image pairs were selected from these screened tiles. Based on (1), about 8.3% of the clear pixel pairs were removed from the selected all image pairs. These removed clear pixel pairs with the surface reflectance values differences which exceeded the threshold contained possible sudden changes in land cover types or atmospheric conditions. And the remaining pixel pairs were the available pixel pairs.

Based on the method described in Section III-A, three HLS tiles (HLS IDs: 23LMG, 17RKQ, and 50SMF) covering a spatially heterogeneous landscape were selected to determine the optimal number of pixel samples for pixel sampling. The three HLS tiles have different land cover type percentage as shown in Table III.

We computed the mean and variance in the land cover type percentage for all pixels and sample pixels across the three test tiles. The summarized statistical results in Table IV indicate that the land cover type percentages for the sample with 1000 pixels closely aligned with the entire image and exhibited minimal change as the number of samples increased. Therefore, we opted to set the number of pixel samples as 1000 pixels for each selected image pairs. In Table IV, the large variances in the cropland, forest, grassland, impervious

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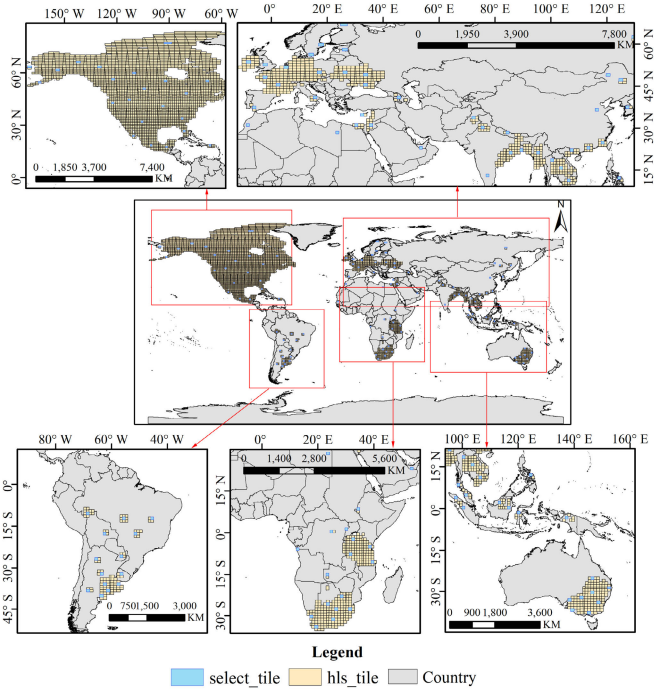


Fig. 4. Distribution of global HLS tiles (orange boxes) and selected HLS tiles (blue boxes).

TABLE III
LAND COVER TYPE PERCENTAGE OF THE THREE HLS TILES

Land cover types (%)	HLS ID		
	23LMG	17RKQ	50SMF
Cropland	20.7	18.4	77.64
Forest	6.87	55.79	5.05
Grassland	16.15	23.23	2.19
Shrubland	55.2	0.01	0.71
Wetland	0.05	0.71	0.06
Water	0.03	0.84	1.29
Tundra	0.0	0.0	0
Impervious surface	0.93	0.97	12.89
Bareland	0.08	0.04	0.16
Snow/Ice	0.0	0.0	0

surface, and shrubland are attributed to the significant differences in the land cover type percentage in the three tiles (see Table III). Additionally, for selected images in cloudy areas where obtaining sufficient samples was challenging, we included all available pixels as samples. Finally, from the selected HLS image pairs, we obtained 4 94 230 available pixel pairs in each matching band to construct the sample library.

B. Spectral Harmonization Model From MA-MP

When we build the spectral harmonization model, to ensure that the training data was as representative as the sample library, we randomly sampled the pixels from the sample library and compared the difference percentage of mean and variance between the sampling results with the sample library.

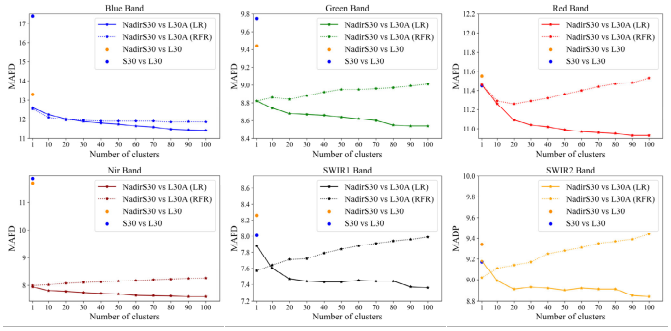


Fig. 5. Performance evaluation results of band-pass adjustment model with different number of cluster sets. The MAPDs of L30 observations and NadirS30 observations or L30 observations and S30 observations were represented by orange dots and blue dots, respectively. The spectral differences between NadirS30 observations and L30 observations after band-pass adjustment based on MA-MP (used LR) were represented by solid lines and based on RFR were represented by dotted lines, respectively.

We sampled the NadirS30 pixels from the sample library and gradually increased the percentage of samples from 10% to 90%. The values of the difference percentage of the mean and variance were expanded 1000 times in magnitude to determine the optimal percentage of samples. To avoid the uncertainty caused by only one random sampling, we sampled 100 times at each percentage and took the average of the mean and variance of 100 times sampling results as the final results of participation in the difference percentage calculation. The statistical results are shown in Table V.

In this study, we empirically considered that the difference percentage between sampling results and the sample library was significant when the difference percentage exceeded 10% in mean or exceeded 30% in variance (the significant differences are shown in bold in Table V). The results showed that significant differences in means may occur when the percentage of samples is less than 60%, and significant differences in variances may occur when the percentage of samples is less than 70%. To balance the representativeness of the training data and the sample size of the validation data, we randomly sampled 70% of the pixels from the sample library as training data and the remaining 30% as validation data.

We used training data for getting clusters and built the spectral harmonization model based on the methods described in Sections III-B and III-C, and used validation data to evaluate model performance based on the method described in Section III-E. The number of clusters was changed from 1 to 100, in which a single cluster (i.e., the number of clusters is 1) means the sensor-specific spectrum was not subdivided into subclusters at all. This case corresponds to building the band-pass adjustment model based on GLR. The evaluation results for the band-pass adjustment model and missing band prediction model, with different numbers of cluster sets, are shown in Figs. 5 and 6, respectively. When counting the spectral difference between L30 observations and NadirS30 observations or between L30 observations and S30 observations, the number of clusters was set to 1.

For the band-pass adjustment (see Fig. 5), the results showed that the MAPDs between L30 observations and NadirS30 observations increased after band-pass adjustment based on GLR in the blue, green, and Nir bands. Specifically, in the blue band, which has the largest MAPD of all the bands, the

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TABLE IV

STATISTICAL RESULTS OF THE MEAN AND VARIANCE OF LAND COVER TYPE PERCENTAGES FOR ALL PIXELS AND SAMPLE PIXELS

Statistical index	Land cover types (%)	Statistical results for different sampling numbers					
		All pixels	Sampled 1000 pixels	Sampled 3000 pixels	Sampled 5000 pixels	Sampled 7000 pixels	Sampled 9000 pixels
Mean	Cropland	38.91	39.01	39.01	38.87	38.94	38.92
	Forest	22.57	22.57	22.50	22.58	22.56	22.52
	Grassland	13.86	13.83	13.84	13.87	13.87	13.89
	Shrubland	18.64	18.64	18.66	18.63	18.65	18.66
	Wetland	0.27	0.26	0.27	0.27	0.27	0.27
	Water	0.72	0.72	0.71	0.72	0.71	0.71
	Tundra	0.00	0.00	0.00	0.00	0.00	0.00
	Impervious surface	4.93	4.89	4.93	4.96	4.91	4.93
	Bareland	0.09	0.09	0.10	0.09	0.09	0.09
	Snow/Ice	0.00	0.00	0.00	0.00	0.00	0.00
Variance	Cropland	1126.14	1123.09	1128.72	1122.52	1126.44	1124.66
	Forest	828.50	828.59	825.50	828.85	830.13	824.16
	Grassland	114.61	113.01	115.32	114.43	114.99	115.16
	Shrubland	1002.60	998.97	1005.88	1001.32	1002.78	1004.97
	Wetland	0.14	0.13	0.13	0.15	0.14	0.15
	Water	0.41	0.40	0.39	0.42	0.42	0.41
	Tundra	0.00	0.00	0.00	0.00	0.00	0.00
	Impervious surface	47.52	46.22	47.28	47.96	47.24	47.48
	Bareland	0.00	0.00	0.00	0.00	0.00	0.00
	Snow/Ice	0.00	0.00	0.00	0.00	0.00	0.00

TABLE V

STATISTICAL RESULTS OF THE DIFFERENCE PERCENTAGE OF MEAN AND VARIANCE BETWEEN ALL PIXELS WITH SAMPLE PIXELS IN THE SAMPLE LIBRARY. NOTE THAT FOR ALL PIXELS IN THE SAMPLE LIBRARY, THE SAMPLING PERCENTAGE WAS 100%. (BOLD NUMBERS INDICATED THAT THE MEAN OR VARIANCE WAS SIGNIFICANTLY DIFFERENT FROM THE SAMPLE LIBRARY)

The difference percentage*1000	Sampling percentage (%)	Statistical results for each band					
		Blue	Green	Red	NIR	SWIR1	SWIR1
Mean	100	0	0	0	0	0	0
	10	**27.29**	**29.73**	**43.02**	7.27	**21.61**	**28.53**
	20	**22.58**	**19.77**	**24.67**	**12.40**	**22.75**	**24.24**
	30	2.36	3.05	1.87	4.61	1.18	3.88
	40	3.14	1.86	**10.29**	**12.00**	7.12	**12.59**
	50	5.30	8.10	**10.99**	3.18	7.20	**10.05**
	60	1.96	1.33	6.78	4.89	3.36	8.04
	70	3.73	5.31	9.99	2.30	6.77	8.64
	80	0.98	1.59	1.40	3.81	4.06	4.82
	90	2.75	3.45	3.97	1.27	3.45	3.42
Variance	100	0	0	0	0	0	0
	10	**75.53**	**97.98**	**110.99**	19.91	**93.92**	**117.94**
	20	**88.33**	**44.11**	4.64	**50.12**	**49.38**	**50.69**
	30	4.38	10.86	8.41	14.61	2.89	7.96
	40	**44.31**	28.93	17.57	25.63	2.84	9.22
	50	**33.17**	**53.84**	**54.94**	9.27	**41.31**	**49.06**
	60	5.51	15.34	21.74	6.24	**40.22**	**49.14**
	70	5.04	6.31	13.41	9.37	6.25	4.13
	80	26.05	21.96	9.09	11.01	6.23	1.72
	90	13.75	1.77	6.23	9.75	2.55	1.13

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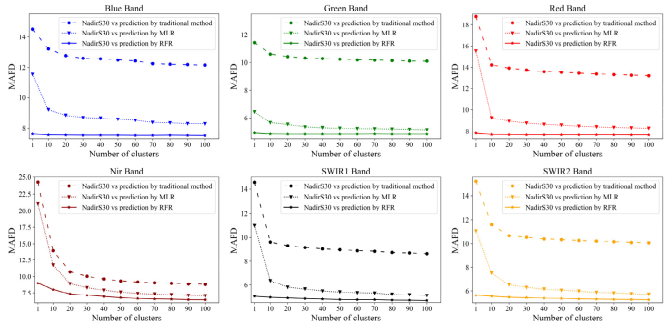


Fig. 6. Performance evaluation results of missing band prediction model with different number of cluster sets.

MAPD increased by nearly 4% after band-pass adjustment. Studies also revealed that in the HLS product, the correlation between Landsat-8 and Sentinel-2 images in the blue band was the worst [23], [29]. Compared to GLR, there were general decreases in MAPDs after band-pass adjustment based on MA-MP (used LR), and the decreases were significant in all bands. For example, in the Nir band, the MAPD between L30 observations and NadirS30 observations increased from 11.69% to 11.85% after using GLR (adjust NadirS30 to L30) and decreased to 7.81% after band-pass adjustment based on MA-MP (adjust L30 to NadirS30) by clustering the training data into only ten clusters, which was a decrease of over 30% compared to the spectral difference before band-pass adjustment. Fig. 5 also indicated that the performance of the band-pass adjustment model from MA-MP was generally better as the number of clusters increased and stabilized when the number of clusters increased to a certain level. Specifically, the MAPDs after band-pass adjustment based on MA-MP decreased as the number of clusters increased in six selected bands and became stable after the number of clusters reached 90 in visible bands, 60 in Nir bands, and 90 in two short-wave infrared bands. Notably, the number of clusters was definitely greater than the number of land cover types. There may be two main reasons for this. First, the samples were from the global scale and therefore the same object may have different spectrums. Additionally, the samples from the same location may be selected from different seasons, so there may be spectrum differences between them.

We replaced the LR with RFR in MA-MP and the performance of the band-pass adjustment model built by these two methods was compared. According to our tests, we used RFR with 250 trees, which represented a good compromise between band-pass adjustment accuracy and processing speed. The findings revealed that the band-pass adjustment results with RFR generally outperformed those with LR only when the number of clusters was set to 1. This discrepancy may arise because LR did not consider the effect of different landscape properties at this stage, whereas RFR, with a sufficiently large number of trees, effectively accommodated the spectral variability within the input data [30]. Unlike the performance of the band-pass adjustment model based on LR, Fig. 5 shows that except for the blue band, the performances of the band-pass adjustment model based on RFR were generally worse as the number of clusters increased. And when the MAPDs stabilized, the MAPD values from band-pass adjustment model based on LR were lower than that of band-pass adjustment model based on RFR in all bands. This may imply that LR models are suitable for building band-pass adjustment models when the effect of different landscape properties is considered, which was consistent with the findings of existing studies [23], [30].

Based on the statistical results in Fig. 5, the MAPDs of the band-pass adjustment results were stable at cluster numbers up to 100 for all selected bands. Therefore, we set the number of clusters to 100 to cluster the training data and then built the band-pass adjustment models that used LR for each cluster, i.e., a total of 100 band-pass adjustment models were obtained in each matching band.

For the missing band prediction, we compared the performance of the missing band prediction model from MA-MP using MLR and RFR (using 250 trees) and the performance from the traditional method [30], which built the MLR model between the NadirS30 observation in the prediction band and L30 observation in the remaining five bands. Fig. 6 illustrates that when using MLR to build the missing band prediction model, the trend of curves depicting prediction accuracy as a function of the number of clusters remains consistent for missing band prediction models built using both MA-MP and the traditional method. Nevertheless, the models built with MA-MP outperformed those built using the traditional method. This superiority arises from the fact that the original spectral differences between NadirS30 and L30 observations are transmitted to the missing band prediction model built with the traditional method, resulting in a decline in prediction accuracy.

We also conducted a performance comparison of the missing band prediction model from MA-MP using MLR with RFR, and the models using RFR consistently outperformed those using MLR for different numbers of clusters in all bands (see Fig. 6). This suggests that the relationship between the prediction band observations and the remaining band observations in the spectral curve of a single sensor pixel, such as NadirS30, may exhibit nonlinearity. Furthermore, when building the missing band prediction model using RFR, the improvements in prediction accuracy with an increasing number of clusters were generally marginal. The MAPDs of the prediction results from MA-MP using RFR stabilized after the number of clusters reached ten in visible bands and two short-wave infrared bands, and 60 in Nir band. This was also attributed to RFR effectively accommodating the spectral variability within the input data.

As the red-edge band that lacked in Landsat-8 OLI was not corrected for the BRDF effect in NadirS30 observations, we selected a band within the matched six bands, which had undergone BRDF effect correction, as the predicted band. The larger the MAPD values in the performance evaluation results of the missing band prediction model based on RFR in Fig. 6, the lower the model performance in the six bands. Obviously, if robust prediction results can be achieved using the missing band prediction model with the lowest performance among the six bands, it implies that the performance of the missing band prediction model based on MA-MP in the other five bands can also be demonstrated. Therefore, we selected the red band with the highest MAPD values as the predicted band. Consequently, the matched band observations utilized in building the missing band prediction model included blue, green, Nir, and two short-wave infrared bands. Moreover,

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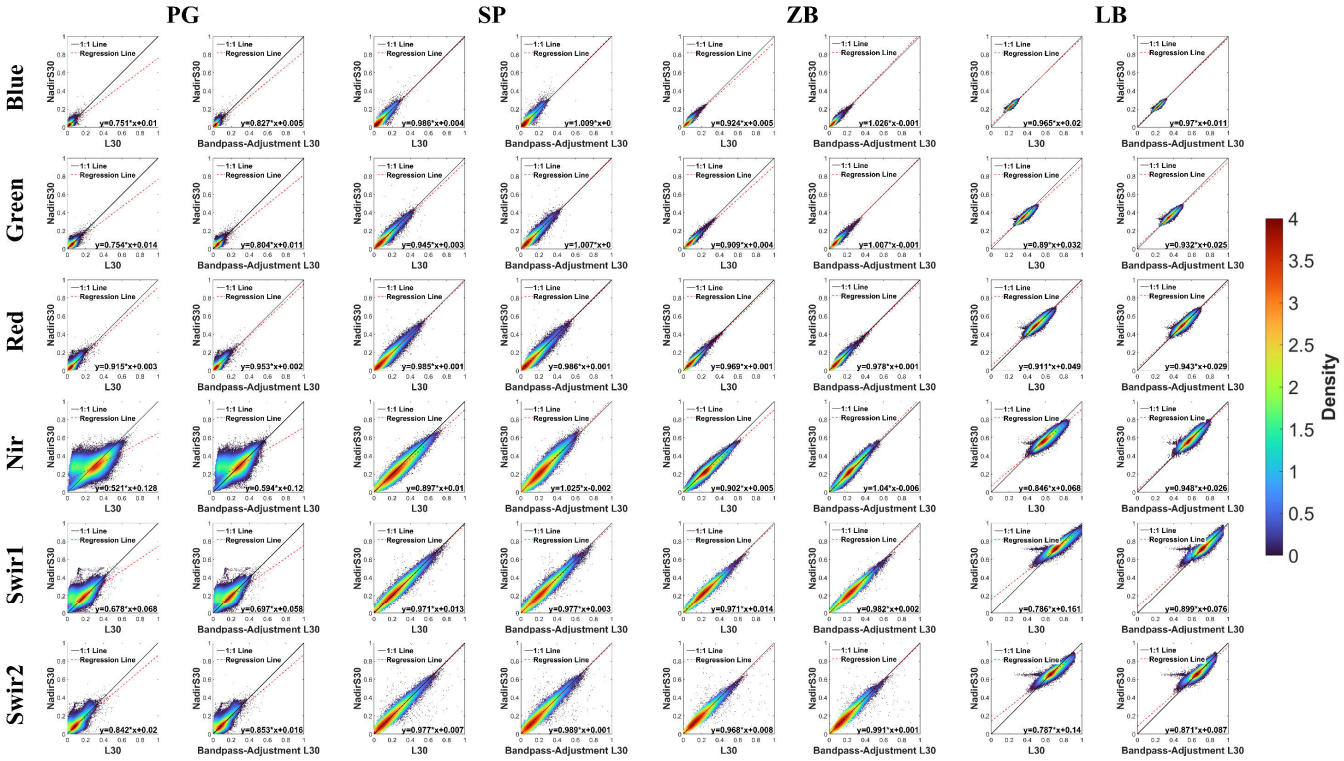


Fig. 7. Scatterplots of NadirS30 images and L30 images before and after the band-pass adjustment for the available pixel pairs at the four sites. The plot colors show the relative frequency of occurrence of similar pixel pairs. The red dotted lines show OLS LR fits of these data.

to ensure the band-pass adjustment accuracy and prevent secondary clustering, we performed missing band prediction at the cluster level based on the clustering results of band-pass adjustment.

C. Reduce Spectral Difference Using the Band-Pass Adjustment Model

At the four selected sites, we implemented band-pass adjustment for L30 images and compared the spectral difference between L30 images before and after band-pass adjustment and NadirS30 images in each matching band for the image pairs at each site.

In the scatterplots of the NadirS30 images and L30 images before and after the band-pass adjustment at the four sites, the red dotted lines showed ordinary least squares (OLS) LR fits of these data (see Fig. 7). The closer the regression line is to the 1:1 line, the closer the NadirS30 images and L30 images are. The results indicate that for all bands, the regression lines were always further away from the 1:1 line for scatterplots of NadirS30 images and L30 images than scatterplots of NadirS30 images and band-pass-adjustment L30 images at the four sites. This means that after band-pass adjustment based on MA-MP, the L30 images were closer to the NadirS30 images in surface reflectance. Especially after band-pass adjustment, the regression lines of scatterplots and 1:1 line almost overlapped for the red band at the PG site, the green band at the ZB site, and the Nir band at the SP and LB sites, which indicated that band-pass-adjustment L30 images closely approximated to NadirS30 images in surface reflectance.

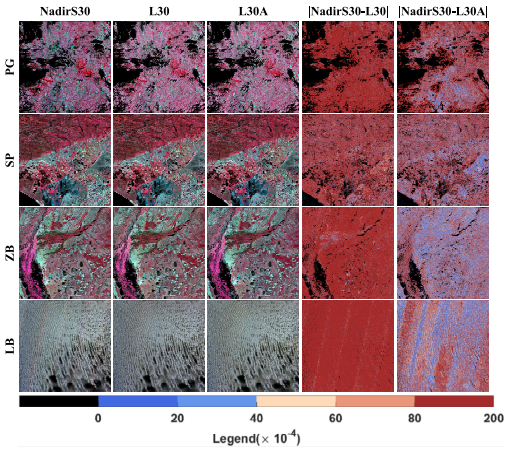


Fig. 8. Standard false-color-composite NadirS30 observations, L30 observations, and the band-pass-adjustment L30 observations (L30A), and their absolute difference images of the Nir band surface reflectance at the four sites. Black areas in the absolute difference images indicate invalid observations.

Fig. 8 illustrates a typical example of band-pass adjustment based on MA-MP for the NadirS30 and L30 images at the four sites. The first three columns present the standard false-color-composite images made out of the Nir, red, and green bands from NadirS30, L30, and the band-pass-adjustment L30 (L30A), respectively. The last two columns depict absolute difference images of surface reflectance in the Nir band between NadirS30 and L30 and between NadirS30 and L30A, respectively. The values of the legend and the segments were set in accordance with existing studies [23]. Obviously, after band-pass adjustment based on MA-MP, the surface

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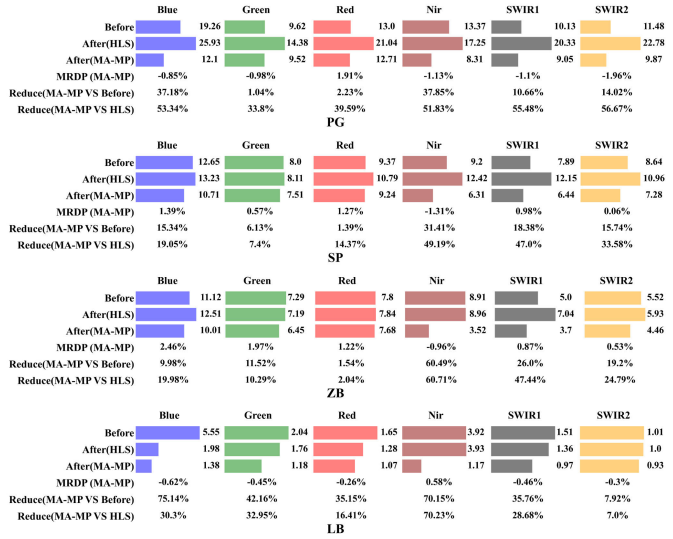


Fig. 9. MAPDs and MRPDs before and after the band-pass adjustment and the reduced percentage of MAPDs based on the GLR (in the HLS product) and MA-MP for six matching bands at the four sites.

reflectance difference between the NadirS30 and L30 images was substantially minimized. Furthermore, the spatial details of the land surface in the L30A standard false-color-composite images remained consistent with those in the L30 images. These results indicated that the MA-MP method not only significantly reduced the spectral difference between NadirS30 and L30 images but was also effective in preserving the spatial patterns and heterogeneity of land surfaces.

We further compared the performance of the band-pass adjustment model from MA-MP with the band-pass adjustment model from GLR. The MAPDs before and after the band-pass adjustment, as well as the reduced percentage of MAPDs based on MA-MP for six matching bands at each site, are shown in Fig. 9. It is clear that at the PG, SP, and ZB sites, there were always bands where the spectral difference increased after band-pass adjustment in the HLS product based on GLR. For example, at the PG and SP sites, the MAPDs after band-pass adjustment in the HLS product increased in all bands. In particular, at the PG site, after band-pass adjustment in HLS, the MAPDs approximately doubled in two short-wave infrared bands compared to before band-pass adjustment. Only at the LB site with a single land cover type, spectral differences can be generally reduced in the HLS product. This may be due to the fact that the global band-pass adjustment model can only describe the overall characteristics of the landscape properties of the samples involved in building the global model, making it challenging for the global model to maintain robust band-pass adjustment results for pixels with different land cover types. When the characteristics of pixels significantly differ from those described by the global model, applying this global band-pass adjustment model to these pixels cannot significantly reduce the spectral differences and may even result in an increase in the spectral differences compared to before band-pass adjustment. Obviously, this situation is more likely to occur in areas with heterogeneous landscapes, such as the PG, SP, and ZB sites. Relatively, the spectral differences between NadirS30 images and L30 images were always reduced for six matching bands at all sites by using the band-pass adjustment model from MA-MP.

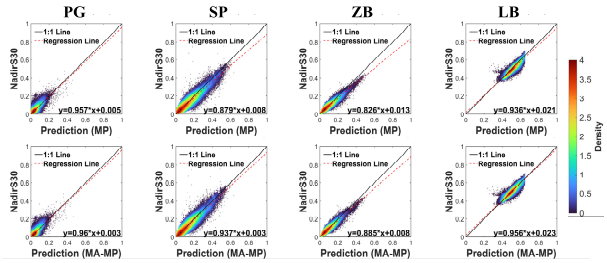


Fig. 10. Scatterplots of NadirS30 images and L30 prediction images based on MA-MP for the available pixel pairs in the red band at the four sites. The plot colors show the relative frequency of occurrence of similar pixel pairs. The red dotted lines show OLS LR fits of these data.

In the two short-wave infrared bands at the PG site, MAPDs were reduced by 10.66% and 14.02% after band-pass adjustment from MA-MP compared to before band-pass adjustment and reduced by 55.48% and 56.67% compared to band-pass adjustment in the HLS product. In addition, in these bands with reduced MAPDs in the HLS product at all sites, the results of band-pass adjustment based on MA-MP were also better than the results in the HLS product. For example, at the LB site, the MAPDs between NadirS30 images and L30 images after band-pass adjustment based on MA-MP were reduced by 30.3% in the blue band, 32.95% in the green band, 16.41% in the red band, 70.23% in the Nir band, 28.68% in the SWIR1 band, and 7.0% in the SWIR2 band compared to band-pass adjustment in the HLS product, respectively. These results demonstrated that, when performing band-pass adjustment, using a band-pass adjustment model that accounts for the effect of different landscape properties is more effective than relying on a global model.

In addition, for six matching bands at all sites, the MRPD values between NadirS30 images and L30 images after band-pass adjustment based on MA-MP ranged from -2% to 2.5%. Over half of the band-pass adjustment results exhibited MRPD values within only $\pm 1\%$ at all sites. Especially at the LB site, the MRPD values were confined to a narrow range of $\pm 0.6\%$. These findings suggest that there is no significant over/underestimation for L30 images after band-pass adjustment based on MA-MP.

D. Prediction Using the Missing Band Prediction Model

We set the red band as the missing band in L30 and predicted it at the four sites based on the missing band prediction model from MA-MP. Using the evaluation method described in Section III-E, we evaluated the accuracy of the prediction results based on MA-MP. Additionally, we compared the performance of the missing band prediction model based on samples before band-pass adjustment (i.e., just MP and not MA-MP) with the missing band prediction model from MA-MP.

Fig. 10 shows the scatterplots of NadirS30 images and L30 prediction images based on MA-MP and MP for the available pixel pairs at the four sites. The plot colors represent the relative frequency of occurrence of similar pixel pairs. The red dotted lines depict OLS LR fits of these data. The results indicate that for all bands, the regression lines were consistently positioned closer to the 1:1 line for scatterplots of NadirS30 images and L30 prediction images from MA-MP

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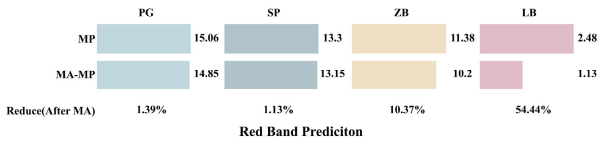


Fig. 11. MAPDs between NadirS30 observation and L30 prediction results used MP and MA-MP in the red band, and the reduced percentage of MAPDs of missing band prediction after band-pass adjustment at the four sites.

compared to scatterplots of NadirS30 images and L30 prediction images from MP at the four sites. This means that when using L30 images after band-pass adjustment based on MA-MP, the prediction results were closer to the NadirS30 observations. Especially at the LB site, when the prediction was based on L30A instead of L30, the regression lines in scatterplots almost overlapped with the 1:1 line.

Fig. 11 displays the MAPDs between the NadirS30 observation and L30 prediction results based on MA-MP and MP, along with the reduced percentage of MAPDs based on MA-MP compared to using only MP, at the four sites. Compared to building a missing band prediction model based on MP, the MAPDs of the L30 prediction results from MA-MP were reduced by 1.39% at the PG site, 1.13% at the SP site, 10.37% at the ZB site, and 54.44% at the LB site, respectively. This demonstrates that the framework in MA-MP, which involves band-pass adjustment before missing band prediction, is reasonable. The missing band prediction model built based on MA-MP not only captures the spectral curve characteristics of various landscape properties but also mitigates the impact of the original spectral differences between L30 and NadirS30 observations in the bands used for building the missing band prediction model. The reason for differences in the prediction accuracy improvement at the four sites was discussed in Section V.

V. Discussion

This study introduces a novel spectral harmonization method, MA-MP. And the experimental results demonstrate its capability to obtain robust spectral harmonization results in various scenes. Building on this foundation, several key insights for improving spectral harmonization model performance can be identified.

First, it is essential to build a spectral harmonization model by using globally representative samples. Fig. 5 shows that even when the number of clusters was set to 1, which was equivalent to building a global model, the band-pass adjustment model from MA-MP still effectively reduced the spectral difference between L30 observations and NadirS30 observations and generally outperformed the results in the HLS product (only similar in the red and SWIR2 bands in Fig. 5). This can be attributed to the fact that the training samples from the constructed sample library ensure the spatial and temporal representativeness necessary for building the band-pass adjustment model. In comparison, the band-pass adjustment model in the HLS product was built based on hyperspectral images that suffered from spatial and temporal discontinuities. Therefore, even if the global model was built, the results of the band-pass adjustment from globally representative sample were superior to those from the HLS product. As depicted in Fig. 9, after band-pass adjustment based on MA-MP, the MAPDs were significantly reduced, and the MRPDs remained at low values in all bands at the four sites. This further indicates that the band-pass adjustment model based on the globally representative sample could maintain robust results in various scenes.

Second, it is crucial to consider the effect of different landscape properties when building spectral harmonization model. As shown in Figs. 5 and 6, a noticeable enhancement in spectral harmonization model performance was observed when the number of clusters exceeded 1. Specifically, when the number of clusters increased from 1 to 10, indicating the consideration of different landscape properties in MA-MP, the magnitude of MAPD decrease was most significant (illustrated by the steepest slope in solid lines in Figs. 5 and 6) compared to other MAPD decreases associated with an increasing number of clusters (such as the number of clusters increasing from 10 to 20). Fig. 9 also shows that a global band-pass adjustment model that cannot account for the effect of different landscape properties was challenging to obtain robust results across various scenes, especially for those with nonhomogeneous land cover types, such as the PG, SP, and ZB site in this study. These outcomes underscore the necessity of accounting for the effect of different landscape properties when building a spectral harmonization model.

Third, the performance of the missing band prediction model was found to be correlated with the degree of differences between sensors in bands involved in the missing band prediction. The experimental results (Figs. 10 and 11) indicate that the accuracy of missing band prediction can be improved by reducing the spectral differences in the bands used to build the prediction model. However, Fig. 11 also highlights that there were differences in the prediction accuracy improvement at the four sites, as seen in the reduction in MAPDs when compared to using only MP. This variability may be attributed to differences in the extent of band-pass adjustment. Based on the results shown in Fig. 9, we computed the average reduced percentage of MAPDs in the blue band, green band, and Nir band at the four sites after band-pass adjustment based on MA-MP compared to before band-pass adjustment. The average reduced percentage was approximately 25.3% at the PG site, about 17.8% at the SP site, around 27.3% at the ZB site, and roughly 62.49% at the LB site. The degrees of the average reduced percentage in band-pass adjustment matched the degrees of reduced percentage of MAPDs in missing band prediction at the four sites. These three bands were selected due to their substantial impact on the accuracy of the missing band prediction model, given their proximity to the prediction band in this study [25]. This suggests that when significant spectral differences exist between sensors in the bands involved in the prediction, it is essential to implement missing band prediction using MA-MP.

Finally, the complete spectral harmonization process should account for both band-pass adjustment in matching bands and prediction in missing bands. Undoubtedly, the spectral differences between different sensors manifest not only in the matching bands but also in the missing bands. Thus, it is imperative to address both types of differences for effective spectral harmonization. The MA-MP framework, which integrates both band-pass adjustment and missing band prediction, constitutes a holistic spectral harmonization

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method. The experimental results demonstrate that, based on MA-MP, we not only significantly reduced the spectral difference between Sentinel-2A and Landsat-8 observations in six matching bands, but also accurately predicted the assumed missing band observations in Landsat-8 images. This suggests that band-pass adjustment and missing band prediction can and should be accommodated in the spectral harmonization process.

VI. Conclusion

This study proposed a novel spectral harmonization method, MA-MP, which aims at minimizing the spectral difference between Landsat-8 and Sentinel-2A observations. The results highlighted three main advantages of MA-MP over existing spectral harmonization methods. First, the global representative samples from the harmonized Landsat Sentinel-2 (HLS) product were used to build the spectral harmonization model of Landsat-8 and Sentinel-2A at the cluster level, ensuring universal applicability in various scenes. The results demonstrated robust spectral harmonization results across four selected study areas with diverse land cover types. Second, in MA-MP, we proposed a novel approach for predicting unilaterally missing bands in Landsat-8 images compared to Sentinel-2A images, with both model training results and experimental results showing superior prediction results compared to current mainstream ideas. Third, the MA-MP provides a complete spectral harmonization framework that incorporates both band-pass adjustment and missing band prediction.

Despite the evident advantages of MA-MP, certain aspects need further exploration in future research. For example, the use of a global fixed BRDF shape in the HLS product for BRDF effect correction may be less effective in local areas, leading to surface reflectance differences due to BRDF effects in the constructed sample library. Future efforts will aim to mitigate these differences caused by BRDF effects in the sample library. Additionally, to enhance cluster accuracy, exploring more cluster features and alternative clustering methods could be considered as the potential improvements.

This study exclusively applied the MA-MP framework to build the spectral harmonization model between Landsat-8 and Sentinel-2A images. Future investigations will assess the performance of MA-MP in building the spectral harmonization models between other sensor images, such as Landsat-8 and Sentinel-2B images, and will use more bands, such as the red-edge band.

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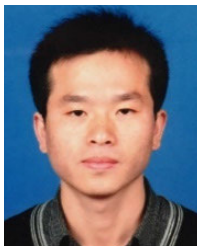
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